

Artificial Intelligence – Comprehensive Notes

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Covering Modules 1.1 – 1.4 | 6-7 Mark Examination Ready

MODULE 1.1: Introduction & History of AI

What is Artificial Intelligence?

Artificial Intelligence (AI) is defined as the simulation of human intelligence processes by machines, especially computer systems. The word 'Artificial' means man-made and 'Intelligence' means the ability to think, learn, and solve problems.

AI processes include:

- **LEARNING** – Acquisition of information and rules for using that information
- **REASONING** – Using rules to reach approximate or definite conclusions
- **SELF-CORRECTION** – Adjusting based on feedback and errors

Particular applications: Expert Systems, Speech Recognition, Machine Vision.

Four Categories of AI Definitions

Russell and Norvig organize AI definitions into four categories:

Category	Focus	Example Definition
Thinking Humanly	Cognitive modeling approach	Automation of activities we associate with human thinking like decision-making, problem solving, learning (Bellman, 1978)
Thinking Rationally	Laws of thought / Logic approach	Study of mental faculties through the use of computational models (Charniak and McDermott, 1985)
Acting Humanly	Turing Test approach	Art of creating machines that perform functions requiring intelligence when performed by people (Kurzweil, 1990)
Acting Rationally	Rational agent approach	Computational Intelligence is the study of the design of intelligent agents (Poole et al., 1998)

Note: The rational agent approach (Acting Rationally) is most favored in modern AI as it is broader than laws of thought and more amenable to scientific development than human-like thinking.

The Turing Test (Acting Humanly)

Proposed by Alan Turing in 1950, the Turing Test was designed to provide a satisfactory operational definition of machine intelligence.

- A computer passes the Turing Test if a human interrogator, after asking written questions, cannot distinguish whether they are talking to a computer or a human.

- For this, a computer needs: Natural Language Processing, Knowledge Representation, Automated Reasoning, Machine Learning.
- The Total Turing Test also requires Computer Vision and Robotics.

Note: Turing said: 'Can machines think?' The test replaces this philosophical question with a practical behavioral one.

Foundations of AI

AI draws from multiple disciplines:

Discipline	Contribution to AI
Philosophy	Concepts of mind as machine; formal rules of thought; idea that thought leads to action
Mathematics	Formal logic, probability theory, computation theory, algorithms
Economics	Decision theory; maximizing expected outcomes; utility theory
Neuroscience	How neurons work; brain similarities and differences from computers
Psychology	Humans and animals as information-processing machines; cognitive models
Computer Engineering	Hardware and software infrastructure enabling AI applications
Linguistics	Language use fits information-processing models; NLP foundations

History of Artificial Intelligence

Key milestones in AI development (from AIMA, Russell & Norvig):

- 1943–1955: Gestation of AI – McCulloch & Pitts proposed neural network model; Turing proposed the imitation game
- 1956: Birth of AI – Dartmouth Conference; John McCarthy coined the term 'Artificial Intelligence'
- 1952–1969: Early enthusiasm – Programs like GPS (General Problem Solver), LISP language created
- 1966–1973: Dose of reality – AI winter; limitations of early systems exposed
- 1969–1979: Knowledge-based systems – MYCIN (medical expert system); knowledge is power
- 1980–present: AI becomes an industry – Expert systems in businesses
- 1986–present: Return of neural networks – Backpropagation algorithm rediscovered
- 1987–present: AI adopts scientific method – More rigorous, empirical approach
- 1995–present: Emergence of intelligent agents – Agents perceive and act in environments
- 2001–present: Very large datasets – Big data, deep learning, modern AI revolution

MODULE 1.2: Intelligent Systems & Components of AI

What is an Intelligent System?

An intelligent system is one that has the ability to:

- Calculate and reason
- Perceive relationships and analogies

- Learn from experience
- Store and retrieve information from memory
- Solve problems and comprehend complex ideas
- Use natural language fluently
- Classify, generalize, and adapt to new situations

Components of Intelligent Systems

Component	Description
Reasoning	Judge and predict; use rules to arrive at conclusions
Learning	Awareness, studying, practising; acquiring new knowledge
Perception	Sensing, organizing, interpreting environmental stimuli
Problem Solving	Decision making; finding solutions through search and planning
Linguistic Intelligence	Comprehend, speak, write; natural language understanding

Difference Between Human and Machine Intelligence

Aspect	Human Intelligence	Machine Intelligence
Pattern vs Rules	Perceives by patterns	Operates by set rules and data
Memory	Stores/recalls by patterns (e.g., 40404040 easy to remember)	Uses searching algorithms
Incomplete Info	Can complete missing/distorted objects	Cannot do this accurately
Context	Understands context and nuance	Needs explicit encoding of context
Common Sense	Natural common sense reasoning	Difficult to encode formally

Components of an AI Program

Three hardware/software/architectural component categories:

1. Hardware Components

- Pattern Matching, Logic Representation, Symbolic Processing
- Numeric Processing, Problem Solving, Heuristic Search
- Natural Language Processing, Knowledge Representation
- Expert Systems, Neural Networks, Learning, Planning

2. Software Components

- Machine Language, Assembly language, High Level Language
- LISP Language (primary AI language historically), Fourth Generation Language
- Object Oriented Language, Distributed Language, Natural Language

3. Architectural Components

- Uniprocessor, Multiprocessor, Special Purpose Processor
- Array Processor, Vector Processor, Parallel Processor, Distributed Processor

Note: Modern AI uses GPU (Graphics Processing Units) massively for parallel processing of neural networks.

MODULE 1.3: Sub-areas of AI

Major Sub-areas of Artificial Intelligence

AI encompasses many specialized domains:

1. Machine Learning (ML)

Machine Learning is a subset of AI that enables systems to learn and improve from experience without being explicitly programmed.

- Supervised Learning: Learns from labelled training data (e.g., spam classification)
- Unsupervised Learning: Finds hidden patterns in unlabelled data (e.g., clustering)
- Reinforcement Learning: Learns through rewards and penalties (e.g., game playing AI)
- Deep Learning: Multi-layer neural networks for complex pattern recognition

Note: ML is the engine of modern AI – most current AI systems rely on ML techniques.

2. Neural Networks (NN)

Inspired by the biological structure of the human brain; consists of interconnected nodes (neurons) organized in layers.

- Input Layer: Receives raw data
- Hidden Layers: Extract features and representations
- Output Layer: Produces predictions or decisions
- Applications: Image recognition, speech recognition, natural language processing
- Deep Neural Networks: Many hidden layers; enables deep learning

3. Robotics

Robotics involves designing, constructing and operating robots – machines that can perform tasks automatically or semi-autonomously.

- Perception: Sensors (cameras, lidar, microphones) to perceive environment
- Cognition: AI algorithms to plan and make decisions
- Action: Actuators (motors, grippers) to interact with the world
- Applications: Industrial automation, surgical robots, self-driving cars, household robots
- Challenge: Real-world environments are unstructured and unpredictable

4. Fuzzy Logic

Traditional logic is binary (True/False). Fuzzy logic handles imprecise or approximate reasoning where values can be partially true.

- Degree of truth ranges from 0 to 1 (not just 0 or 1)
- Closer to human reasoning which is not always crisp
- Example: 'Tall' is fuzzy – 5'11" is 'somewhat tall'; 6'5" is 'very tall'
- Applications: Washing machines (adaptive cycles), air conditioners (auto temperature), anti-lock brakes

Note: Fuzzy logic is especially useful in control systems where precise mathematical models are difficult.

5. Cognitive Computing

Cognitive computing simulates human thought processes in a computerized model. It uses self-learning algorithms that use data mining, pattern recognition and natural language processing.

- Goal: Create automated IT systems that can solve problems without human assistance
 - Key capabilities: Reasoning under uncertainty, learning, interaction with humans naturally
 - IBM Watson is a famous example of a cognitive computing system
 - Applications: Healthcare diagnostics, financial advisory, customer service
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MODULE 1.4: Applications & Current Trends in AI

Applications of AI

Domain	AI Applications
Gaming	Chess, poker, Go – AI evaluates millions of positions using heuristic knowledge
Natural Language Processing	Machine translation, chatbots, sentiment analysis, voice assistants
Expert Systems	Medical diagnosis (MYCIN), financial advisory, legal reasoning
Vision Systems	Medical imaging, satellite analysis, facial recognition, autonomous vehicles
Speech Recognition	Handles accents, slang, background noise; Siri, Alexa, Google Assistant
Handwriting Recognition	Converting handwritten text to editable digital text
Intelligent Robots	Industrial manufacturing, surgery, logistics, exploration
Healthcare	Disease prediction, drug discovery, robotic surgery, radiology
Finance	Fraud detection, algorithmic trading, credit scoring
Transportation	Self-driving cars, traffic management, route optimization
Education	Personalized learning, automated grading, intelligent tutoring systems

Current Trends in AI – Key Concepts

1. Generative AI

AI that can create new content – text, images, audio, video, code – that did not exist before.

- Uses deep learning models trained on large datasets
- Examples: ChatGPT (text), DALL-E (images), GitHub Copilot (code), Sora (video)
- Underlying models: Large Language Models (LLMs), GANs (Generative Adversarial Networks), Diffusion Models
- Applications: Content creation, code generation, art, drug molecule design

Note: Generative AI represents one of the biggest AI breakthroughs – it moves AI from analysis to creation.

2. Predictive AI

AI that uses historical data and statistical models to forecast future outcomes.

- Uses regression, time-series analysis, classification models
- Examples: Weather forecasting, demand forecasting, predictive maintenance
- Healthcare: Predicting disease risk; Finance: Stock price prediction, credit risk
- Works best when historical patterns reliably continue into the future

3. Discriminative AI

AI that learns to distinguish between different classes or categories of data.

- Focuses on learning the boundary between classes (unlike generative AI that models data distribution)
- Examples: Email spam vs. not-spam, tumor vs. healthy tissue, cat vs. dog classifier
- Models: Support Vector Machines (SVM), logistic regression, modern deep classifiers
- High accuracy in well-defined classification tasks

4. Descriptive AI

AI that analyzes historical data to describe what has happened and identify patterns.

- Answers the question: 'What happened?'
- Uses data analytics, data mining, business intelligence
- Examples: Sales dashboards, customer behavior analysis, anomaly detection in logs
- Foundation for more advanced predictive and prescriptive AI

5. Prescriptive AI

AI that goes beyond prediction to recommend actions – it answers 'What should we do?'

- Combines prediction with optimization algorithms
- Examples: Google Maps route optimization, supply chain optimization, treatment recommendation in hospitals
- Uses techniques: Linear programming, reinforcement learning, simulation

Note: The four AI types form a hierarchy: Descriptive → Predictive → Prescriptive → Generative.

6. Adaptive AI

AI that continuously learns and updates its behavior in real-time based on new data and changing environments.

- Unlike static models, adaptive AI evolves after deployment
- Uses online learning, reinforcement learning, transfer learning
- Examples: Recommendation systems (Netflix, Spotify), fraud detection systems that adapt to new fraud patterns
- Key for applications in dynamic environments

7. Interactive AI

AI designed to interact naturally with humans through conversation, gesture, or other interfaces.

- Uses NLP, speech recognition, dialogue management
- Examples: Conversational agents (ChatGPT, Alexa), social robots, interactive educational tutors
- Goal: Human-like natural interaction
- Challenges: Understanding context, sarcasm, ambiguity; maintaining multi-turn conversation

8. Cognitive AI

AI that attempts to mimic human cognitive processes including reasoning, problem-solving, learning, and perception.

- Combines ML, NLP, vision, reasoning and knowledge representation

- IBM Watson is a pioneering example
- Applications: Healthcare diagnosis, legal research, complex Q&A systems
- Strives for general intelligence rather than narrow task performance

9. Explainable AI (XAI)

AI systems whose outputs, decisions and internal workings can be understood and interpreted by humans.

- Addresses the 'black box' problem in deep learning
- Essential for trust, accountability and regulatory compliance

Key XAI Techniques:

Technique	Description
Feature Importance	Identifies which input features most influenced a prediction
LIME	Local Interpretable Model-agnostic Explanations – creates local approximation around a prediction
SHAP	SHapley Additive exPlanations – assigns each feature a contribution value based on game theory
Counterfactual	Explains decisions by showing what would need to change to get a different outcome
Rule-based	Produces if-then rules describing decision logic
Tree SHAP	Efficient SHAP for tree-based models like Random Forest, XGBoost
Deep SHAP	Backpropagates Shapley values through neural networks

Note: Why XAI matters: Detects bias, enables accountability, allows contestability, improves trust, and is legally required in many domains (e.g., GDPR in Europe).

10. Hybrid AI

Combines multiple AI paradigms – symbolic AI and machine learning – to leverage the strengths of each.

- Symbolic AI: Rule-based, explainable, good at reasoning
- Neural AI: Pattern recognition, learns from data, handles unstructured data
- Hybrid: Combines both – e.g., neural network + knowledge graph
- Examples: AlphaGo (deep learning + Monte Carlo tree search), neuro-symbolic AI
- Benefit: More robust, explainable and generalizable than either alone

AI Ethics and Fairness

Why Ethics in AI?

AI is no longer confined to research labs – it influences real human decisions in hiring, lending, healthcare, criminal justice, and governance. Errors can affect lives and fundamental rights.

- Technical success does NOT equal ethical success
- Who is responsible when AI fails? Developer, organization, user, or government?
- Ethics defines accountability beyond code

- Real examples: Hiring AI rejecting female candidates; criminal justice risk scores (COMPAS); recommendation systems amplifying harmful content

Core Ethical Principles of AI

Principle	Explanation & Example
Fairness	AI must not discriminate against individuals or groups; must provide equal treatment. E.g., hiring AI ensures equal selection chances for men and women.
Accountability	Humans/organizations must be responsible for AI decisions. AI must be auditable and contestable; human-in-the-loop is essential. E.g., bank takes responsibility for wrong loan rejection by AI.
Transparency	AI processes and decision-making should be open and understandable. Users should know AI is being used. Transparency ≠ revealing source code. E.g., user informed credit approval was decided by AI.
Privacy & Consent	Personal data must be collected, stored, and used with user consent and protection. Requires lawful data collection, explicit consent, data minimization. E.g., health app asks consent before collecting medical data.
Safety & Reliability	AI must operate correctly, securely, and consistently without causing harm. Must handle edge cases. Critical in healthcare and transport. E.g., self-driving car safely stops when sensors detect obstacle.
Explainability	AI decisions must be explainable in a way humans can understand. Essential for trust, detecting bias, enabling contestability. E.g., loan AI explains rejection due to low credit score.
Human Responsibility	Humans must oversee AI systems and make final decisions, especially in critical cases. Appeals and redress mechanisms required.

Types of Bias in AI

AI bias is one of the most critical ethical challenges. Bias can enter at every stage of the AI pipeline:

Type of Bias	Description & Example
Data Bias	Bias in training data due to imbalance or missing data. E.g., facial recognition trained on light-skinned faces fails on dark-skinned faces.
Sampling Bias	Training data doesn't represent real-world population. E.g., medical AI trained on urban hospital data fails for rural patients.
Algorithmic Bias	Bias from model design, features, or optimization goals. E.g., loan system favoring high-income groups to maximize profit.
Historical Bias	AI learns and reinforces existing societal inequalities. E.g., predictive policing targets same communities repeatedly.
Labeling Bias	Subjective or inconsistent human annotations. E.g., emotion detection labels vary by annotator culture.
Measurement Bias	Incorrect or incomplete measurement methods. E.g., using arrest records as proxy for crime rate.

Confirmation Bias	Developers favor outcomes confirming their assumptions. E.g., ignoring model errors contradicting expected results.
Proxy Bias	Indirect features substitute for sensitive attributes. E.g., ZIP code used as proxy for race in loan approvals.
Deployment Bias	AI used in different context than trained. E.g., speech recognition trained on US accents fails on Indian accents.
Temporal Bias	Data becomes outdated. E.g., job AI using pre-pandemic employment data.

Indian Context – AI Ethics Frameworks

- NITI Aayog – Responsible AI for All: India's national AI strategy emphasizing inclusive, ethical AI
- MeitY – IndiaAI Mission: Ministry of Electronics and IT driving responsible AI deployment
- DPDP Act 2023 – Digital Personal Data Protection Act: India's data privacy law governing AI data use

Key considerations for India:

- Large-scale public AI deployment (Aadhaar, DigiLocker, CoWIN) – high impact on 1.4 billion citizens
- Linguistic and cultural diversity requires culturally sensitive and multilingual AI
- Explainability is critical for inclusion – rural and low-literacy populations need understandable AI

Ethics + Explainability: The Connection

Ethics and Explainability are two sides of the same coin in trustworthy AI:

Ethics	Explainability
Defines WHAT should be done (values, principles, fairness)	Shows WHY it was done (reasons, logic, factors)
Sets the goals and constraints	Verifies those goals were actually met
Answers: Is this AI system behaving fairly?	Answers: How did the AI system arrive at this decision?
Example: Fairness policy in hiring AI	Example: SHAP values showing which features drove rejection

Both are required for trustworthy AI. A system can be ethical in design but black-box in practice; or it can be explainable but still encode bias.

Role of Future Engineers

- Question data and assumptions before training any model
 - Avoid blind trust in AI outputs – always validate and test
 - Design human-centric AI that keeps humans in control
 - Recommend and implement governance strategies
 - Apply ethical theories to technical systems during design, not after deployment
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AI Problems & AI Techniques (Supplementary)

AI Problem Domains (Rich & Knight Classification)

Mundane Tasks

- Perception: Vision, Speech
- Natural Language: Understanding, Generation, Translation
- Commonsense reasoning, Robot control

Formal Tasks

- Games: Chess, Backgammon, Checkers, Go
- Mathematics: Geometry, Logic, Integral calculus, Proving properties of programs

Expert Tasks

- Engineering: Design, Fault finding, Manufacturing planning
- Scientific analysis, Medical diagnosis, Financial analysis

AI Technique Requirements

A good AI technique is a method that exploits knowledge represented in such a way that:

- Knowledge captures generalizations – situations sharing important properties are grouped together
- Understood by people who must provide it – humans can verify, correct, and extend the knowledge
- Easily modified to correct errors and reflect changes in the world
- Can be used in many situations even if not totally accurate or complete
- Helps narrow the range of possibilities that must be considered

What AI Cannot Do (Yet)

- True commonsense reasoning at human level
- Automated scientific discovery (hypothesis generation and testing)
- Fully reliable autonomous medical diagnosis in all cases
- Safe and reliable driverless cars in all conditions
- Creative writing and art with genuine understanding
- Exercising free will or genuine consciousness

Note: The gap between current AI capabilities and human-level general intelligence (AGI) remains substantial. Current AI excels at narrow tasks but struggles with generalization.

Quick Revision – Key Definitions

Term	One-Line Definition
Artificial Intelligence	Simulation of human intelligence processes by machines
Machine Learning	AI subset that learns from experience without explicit programming
Deep Learning	ML using multi-layer neural networks for complex pattern recognition
Neural Network	Computing model inspired by biological neurons; processes data in layers

Fuzzy Logic	Logic handling partial truth; values between 0 and 1
Cognitive Computing	Systems that simulate human thought processes for decision support
Generative AI	AI that creates new content (text, images, code, etc.)
Predictive AI	AI that forecasts future outcomes from historical data
Discriminative AI	AI that classifies or distinguishes between categories
Prescriptive AI	AI that recommends optimal actions
Explainable AI (XAI)	AI whose decisions are understandable by humans
Hybrid AI	Combination of symbolic and neural AI approaches
AI Bias	Systematic errors in AI arising from flawed data or design
LIME	Local Interpretable Model-agnostic Explanations – local explanation technique
SHAP	SHapley Additive exPlanations – contribution of each feature to prediction
Turing Test	Test of machine intelligence via indistinguishable conversation with humans
Rational Agent	Agent that acts to achieve the best expected outcome
DPDP Act 2023	India's Digital Personal Data Protection Act governing AI data use

Prepared from: Lecture slides (Ms. Swati Mali) | AIMA Textbook (Russell & Norvig) | Ethical & Explainable AI Course Material